Applied Energy 410 (2026) 127599

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen

Short-term forecasting of energy production and consumption using extreme learning machine: A comprehensive MIMO based ELM approach

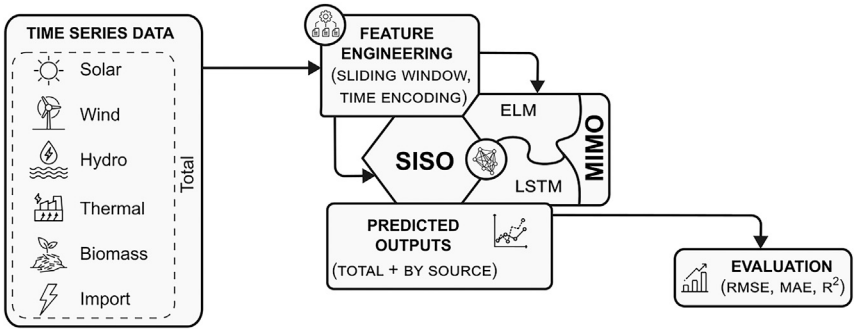
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HIGHLIGHTS

- A novel Extreme Learning Machine-based Multi-Input Multi-Output (MIMO) framework for short-term energy forecasting is proposed.
- The model accurately predicts energy outputs from multiple renewable and non-renewable sources, achieving high accuracy up to five hours ahead.
- Performance is evaluated using normalized RMSE, MAE, and R^2, demonstrating significant improvements over the persistence SISO model and deep learning-based LSTM approaches.
- Accurate forecasting can be a powerful decision-support tool, optimizing dispatch and aiding renewable integration within operational constraints.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Short-term energy forecasting
Extreme learning machine
Multi-source energy systems
MIMO
SISO
Grid integration

ABSTRACT

A multiple-input multiple-output (MIMO) extreme learning machine (ELM) is introduced for short-term forecasting of seven grid variables in Corsica (France): total demand and generation from solar, wind, hydropower, thermal, bioenergy, and imports. Based on six years of hourly data, the model integrates sliding windows and cyclic time encodings to handle non-stationarity and seasonal effects without heavy preprocessing. At a 1-hour horizon, solar and thermal achieve nRMSE of 0.179 and 0.051 with $R^2 > 0.98$, while total demand forecasts remain reliable up to 5 h ahead. Wind and bioenergy remain challenging due to high intrinsic variability, but overall accuracy is robust across sources. Compared with persistence and an LSTM configured under realistic tuning budgets, MIMO – ELM consistently improves skill, offering small but stable gains over Single-Input Single-Output models (SISO). Beyond accuracy, the closed-form solution ensures fast training and suitability for real-time updates, enabling potential use in online learning contexts. A key advantage of the MIMO formulation is internal coherence

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https://doi.org/10.1016/j.apenergy.2026.127599
Received 20 February 2025; Received in revised form 26 December 2025; Accepted 21 February 2026
Available online 26 February 2026
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between aggregate demand and its components, an important requirement for operators. The methodology adapts to local constraints such as grid characteristics, resource availability, and market structures, ensuring transferability beyond the Corsican case. The study shows that a parsimonious approach such as MIMO – ELM can deliver forecasts that are accurate, coherent, and computationally efficient, providing a practical decision-support tool for energy management and renewable integration.

1. Introduction

The integration of renewable energy (RE) sources, such as solar PV and wind, is important for reducing greenhouse gas emissions and mitigating climate change [1–3]. However, their weather-dependency introduces variability in power supply, posing challenges for grids originally designed for stable fossil fuel or nuclear generation [4]. Ensuring grid stability and balancing supply and demand remain key challenges in high-RE systems [1], making accurate forecasting essential for grid management and energy storage optimization [5]. Despite the inherent variability of wind and solar PV [6], machine learning advances now enable more precise short-term energy predictions through large datasets and advanced computational methods [7].

1.1. Motivation for short-term forecasting in multi-source energy systems

The increasing complexity of multi-source grids necessitates real-time adjustments to balance supply and demand, minimize fossil fuel reliance, and improve grid stability. Effective decision-making requires not only accurate predictions but also a broader assessment of resource availability, generation constraints, and transmission delays. Forecasting should be viewed as a decision-support tool for system operators, integrated into broader optimization frameworks that consider real-time constraints and flexibility mechanisms. The increasing role of self-consumption in distributed photovoltaic (PV) systems further complicates demand forecasting. As more energy is consumed directly at the production site, PV capacity translates less directly into injected power, instead reducing the total power drawn from the grid. This shift necessitates integrated forecasting tools that account for both generation and consumption, a need that will only intensify with rising self-consumption levels. Traditional statistical methods [8] often struggle with the nonlinear interactions within complex multi-source energy systems. Machine learning techniques, such as Extreme Learning Machines (ELM), offer improved accuracy by integrating meteorological data, historical trends, and grid conditions [9–11].

A Multiple-Input Multiple-Output (MIMO) approach extends this capability by simultaneously predicting multiple energy variables (e.g., solar irradiance, wind speed, demand, and grid load). This holistic approach enables better coordination of generation, storage, and dispatch, leading to more efficient and reliable grid operation by capturing the complex interdependencies between these variables. While thermal power output is influenced by operational decisions, fuel availability, and technical constraints, forecasting its potential evolution remains valuable, especially when combined with information on hydropower and storage. This allows for better anticipation of critical periods and cost fluctuations. Transmission system operators also rely on accurate forecasts to anticipate power flows. Inaccurate predictions can lead to costly imbalances, impacting network stability, a key concern for these operators. Forecasts can also serve as a substitute for real-time measurements in the (many) cases where such data are unavailable. Beyond operational concerns, the increasing digitalization of energy systems introduces cybersecurity risks. False data injection attacks and manipulation of measurement signals highlight the vulnerability of grid management systems [12,13]. Comparing predicted values with actual grid measurements can help detect anomalies, improving both operational efficiency and security [14,15]. Although several studies employ multi-input/multi-output models for load, residual load, or renewable generation forecasting [16–18], these approaches typically address loosely coupled multi-output problems or isolated energy variables. This differs from the integrated and structurally coherent multi-energy MIMO framework considered here. To our knowledge, few existing studies address a truly multi-energy forecasting setting in which heterogeneous sources (thermal, hydro, solar, wind, imports) are jointly modeled within a unified optimization framework using real operational data.

As a result, while MIMO forecasting is one component of effective grid management, its ability to integrate diverse data sources, account for system complexities, and support real-time decision-making makes it an important tool for navigating the technical, economic, and increasingly decentralized challenges of the evolving energy landscape. This includes not only traditional grid operations but also the efficient management of distributed resources like microgrids, where forecasting plays a vital role in optimizing energy use and maximizing the benefits of local generation and storage. Specifically, for interconnected microgrids, forecasting power limitations and electricity prices on the main grid is challenging for optimizing energy trading. A forecasting horizon of several hours enables better management of storage, load, and potentially even the scheduling of generation resources, allowing microgrids to maximize economic benefits or minimize their environmental impact.

In short-term multi-energy forecasting, a SISO (Single-Input-Single-Output) design treats each energy stream independently, preventing the exploitation of cross-variable dependencies and offering no guarantee of internal coherence between component and aggregate forecasts. Recurrent architectures such as LSTM models remain effective but involve high computational costs, long training times, and limited scalability for operational settings requiring frequent updates. Hybrid AI-based schemes may improve accuracy but generally introduce additional complexity at the expense of interpretability and reproducibility. A compact MIMO formulation addresses these limitations by jointly forecasting all variables within a unified linear-output structure, enabling the learning of compensatory interactions between sources while preserving the computational efficiency and transparency characteristic of extreme learning machines.

The main methodological contribution of this study is the introduction of a unified multi-output ELM formulation ensuring internal coherence between individual energy sources and their aggregate total. Unlike prior ELM or MIMO applications treating each output independently, the proposed model performs a single closed-form regression that simultaneously forecasts seven correlated variables. This design captures cross-compensations between sources while reducing computational costs and preserving analytical tractability.

The structure of this paper is as follows: Section 2 reviews previous research on short-term energy forecasting, focusing on machine learning approaches, in particular the use of ELM in energy systems. Based on the energy problem and the relevant ad-hoc literature, Section 3 introduces and outlines the objectives of this study. Section 4 presents the methodology, detailing the ELM-based forecasting model, preprocessing, architecture and performance measures. Section 5 describes the experimental setup, including the dataset, evaluation framework and reference models, followed by a presentation of the results, discussing the accuracy, computational efficiency and implications of real-time forecasting in multi-source energy networks, illustrated by Corsica. Section 6 concludes by summarizing the main results, addressing limitations and suggesting directions for future research.

2. Related works

This section reviews the main advancements in RE forecasting. Various methodologies are discussed, underscoring their merits and

drawbacks, while emphasizing the benefits of ELM over conventional techniques.

2.1. Energy forecasting approaches

Energy forecasting plays an important role in balancing supply and demand in modern electricity systems, especially with the increasing integration of RE sources. Recent advances in machine learning and, in general, the development of artificial intelligence (AI) have revolutionized the ability to accurately predict RE and non-RE production. [19–21]. Techniques such as Support Vector Regression (SVR), deep learning models, and hybrid AI approaches have shown great potential in capturing complex patterns in energy datasets. In [22] neural networks are used to capture seasonal patterns and abrupt variations in energy consumption data, which is especially useful for managing intermittent energy sources that exhibit high variability.

In recent research, Saglam et al. [23] introduced a new method for predicting electricity demand in Turkey. The authors used Particle Swarm Optimization (PSO) with Support Vector Regression (SVR). This hybrid PSO – SVR model demonstrated significant improvements, illustrating how optimization techniques can enhance machine learning applications in energy forecasting. Along similar lines, Ardabili et al. [24] conducted an in-depth review of deep learning applications for forecasting energy demand. Their work emphasizes the advantages of multimodal AI systems in managing both RE and non-RE sources, offering insights into the evolving role of AI in energy systems management.

In addition, Motwakel et al. [25] proposed a deep learning-based model that combines SVR, Lasso and Ridge regression to predict non-RE consumption, achieving state-of-the-art accuracy in multimodal forecasting scenarios. On the RE side, Ding et al. [26] integrated long-term memory models (LSTM) and SVR to improve solar power generation forecasting, demonstrating the effectiveness of hybrid models in reducing prediction errors. In addition, Revathi [27] conducted a wide-ranging investigation of AI-driven techniques for RE production, focusing on deep learning architectures, and highlighting their potential for optimizing both production and demand management.

In summary, existing forecasting approaches present well-identified strengths and weaknesses. Classical statistical models (e.g., ARIMA, regression) are simple and interpretable but often inadequate to capture nonlinear dynamics. Machine learning models such as support vector regression or random forests better capture nonlinearities and interactions, yet require extensive hyperparameter tuning and large datasets. Deep learning architectures (LSTM, TFT, TSMixer, etc.) achieve strong accuracy in many studies, particularly with exogenous variables, but they are computationally demanding, less transparent, and sometimes prone to overfitting. A recurrent limitation across the literature is the difficulty of generalizing models to different contexts and the lack of explicit consideration for multi-output coherence.

2.2. Advantages of ELM in energy forecasting

ELM shows significant potential in energy demand and source forecasting, particularly when applied to RE sources. The rapid learning capacity and generalization performance of ELM have been widely recognized in recent studies. Abd El-Aziz et al. [28] demonstrated that ELM improves the accuracy of consumption forecasting for RE sources by integrating deep learning models for hybrid RE systems. Recently, Li et al. [29] also took up the challenge by using tools like ELM and SVR to understand and predict the carbon emissions. They found that ELM, a model simple in spirit but powerful in scope, had an advantage. Especially in rugged places where coal and oil still have power, ELM proved relevant.

The same year, Sibtain et al. [30] joined the topic. They designed a multi-stage deep learning machine, a powerful ally for predicting wind vagaries. For those working on RE, knowing the next value of the wind speed can be complicated or even impossible. Their results highlighted a truth: ELM has, time and again, outperformed older methods, proving itself in the wild terrain of green energy and fossil fuel dependency.

In line with the no free lunch theorem, no forecasting algorithm is universally optimal. Extreme Learning Machine is deliberately favored because it provides a closed-form solution, very fast training, and full interpretability, in contrast with deep learning models often perceived as black boxes. Following the parsimony principle, the simplest approach that generalizes well is often the most robust for operational use. ELM allows us to imagine a future in which energy demands and resources are easily predictable, however complex the challenge may be. To our knowledge, no studies or tools have been proposed in the literature for the simultaneous forecasting of all production sources and electricity consumption within a region. This study positions itself as a pioneer in this field.

3. Objective of the study

The objective of this work is to propose a streamlined and efficient methodology for short-term energy forecasting without relying on data stationarity. Energy systems, including both RE and non-RE sources as well as electricity demand, inherently exhibit non-stationary behaviour driven by seasonal variability, maintenance operations, and operational constraints. However, many conventional forecasting approaches assume stationarity [31], an assumption that is often violated in energy systems where cyclic and dynamic patterns dominate. To address this limitation, our approach integrates temporal cues and periodic patterns directly within the model structure, allowing it to adapt dynamically without requiring explicit stationarization. This preserves the intrinsic characteristics of the data while improving predictive performance. While an alternative strategy could rely on training separate models for each hour of the day, the proposed approach captures these temporal variations within a unified framework, ensuring coherence across time. Beyond predictive accuracy alone, this work deliberately emphasizes operational efficiency, robustness, and coherence across multiple energy variables.

In grid operation, forecasting models must remain computationally efficient, stable over time, and easily deployable under real-time constraints. In this context, simplicity is not a limitation but a methodological choice: overly complex models may offer marginal accuracy improvements at the cost of interpretability, reproducibility, and operational reliability. The proposed MIMO–ELM framework is therefore not intended to outperform all existing models in absolute terms, but rather to provide a balanced compromise between accuracy, computational cost, and practical usability. As illustrated through the comparison with LSTM, increased architectural complexity does not necessarily translate into improved performance for this class of energy forecasting problems. Instead, a parsimonious formulation can deliver stable and competitive results while ensuring fast training, deterministic convergence, and suitability for real-time deployment. This choice is thus methodological rather than comparative, aiming at operational robustness rather than benchmark dominance.

This study further investigates the respective roles of Single-Input Single-Output (SISO) and Multiple-Input Multiple-Output (MIMO) formulations in forecasting energy production and demand. Other approaches, such as the TSMixer or Temporal Fusion Transformers (TFT), have also demonstrated strong performance in specific contexts, but remain substantially more complex and computationally demanding. The selection of ELM is therefore intentional, reflecting a compromise between accuracy, efficiency, and operational suitability. This design choice is particularly relevant for energy systems, where forecasts must remain coherent across multiple variables and be updated frequently without excessive computational overhead. In this context, the MIMO formulation offers a structural advantage by enforcing internal consistency between individual energy components and their aggregate, a property that cannot be guaranteed when independent models are trained separately.

For these reasons, the comparative analysis focuses on baselines implemented under identical experimental conditions. Models such

as GRU, Transformer, or hybrid ARIMA–ELM architectures (often designed for single-variable forecasting and lacking established multi-energy benchmarks) were not considered, as they would not constitute methodologically comparable references. Unlike previous studies where ELM was mainly applied to single-output problems, this work introduces a genuine multi-output formulation, enabling simultaneous forecasting of multiple energy sources while preserving internal consistency with aggregated totals, which constitutes the main methodological contribution. Based on these methodological choices, the following section evaluates the proposed framework under realistic operational conditions. To assess performance, the models are evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2), allowing a balanced analysis of accuracy, robustness, and interpretability across forecasting horizons.

4. Materials & methods

In this study, time series representing the hourly electric power produced by different sources are collected and analyzed to assess their respective contributions to Corsica overall energy generation. The workflow of the Multiple Input, Multiple Output (MIMO) predictive method is detailed in Appendix A. In machine learning, Single Input, Single Output (SISO) and MIMO are two modeling approaches with distinct trade-offs. SISO ELM trains separate models for each output variable using a single input feature, offering better interpretability and flexibility but increasing computational cost. In contrast, MIMO ELM predicts all outputs simultaneously using multiple inputs, capturing interdependencies between outputs and being more computationally efficient, though potentially sacrificing accuracy for individual predictions. SISO is usually ideal when outputs are independent, while MIMO is preferable when outputs are correlated, improving overall consistency and reducing training time.

4.1. Data description

While such a study could be performed using data from any region, insular data were specifically chosen to address the unique challenges of islands, such as isolated grids, supply-demand imbalances, and seasonal variations.

Limited interconnections and high RE intermittency further complicate grid stability [32]. Small grids with low inertia lead to frequency variability and voltage instability. Limited spatial aggregation results in fluctuating RE generation, sometimes exceeding safety thresholds, requiring curtailments. Seasonal demand variations necessitate flexible power generation, with Corsican electricity demand ranging from 130 MW to 450 MW, depending on the period. Corsica, the smallest and most mountainous island in the western Mediterranean (8680 km^2 & 568 m avg. altitude), had approximately 350,000 inhabitants in 2022 [33]. Tourism, with 3 million annual visitors mainly from May to September, significantly impacts electricity demand. Currently, Corsica electrical system has a total installed capacity of around 1 GW, including 369 MW from thermal sources, 225 MW from hydropower, 233 MWp from solar PV, 18 MW from wind, 2 MW from biogas, 150 MW from imports, and 6 MW from battery storage. The island is partially connected to Italy through the SACOI interconnection (50 MW DC, operational since 1986) and SARCO (100 MW AC, since 2006). However, Corsica can only import electricity due to commercial constraints. Electricity production remains highly dependent on diesel generators, resulting in CO_2 emissions up to 15 times higher than those in mainland France. Around 70% of electricity is still fossil-fuel-based [32,34].

The dataset used in this study includes electricity demand and key energy sources such as solar PV, wind, hydropower, thermal, bioenergy, and imports, covering the six previous years (2018-2023) at hourly resolution, resulting in 368,256 data points, with additional data for leap years. These data, managed by Électricité de France (EDF), the leading electricity provider in France and responsible for grid management in Corsica, are publicly available at https://www.data.corsica/pages/portail/ [35]. The dataset is based on measurements aggregated as average values over hourly time slots (hourly granularity), enabling detailed temporal analysis. While hourly data may be considered coarse compared to the 15-minute resolution adopted by spot markets, this level of temporal granularity remains sufficient to identify production peaks and observe differences in behavior across energy sources. The datasets used in this study already undergo systematic quality control and homogenization [36]. A visual inspection of the six-year series confirms the absence of missing values or anomalous corrections and shows smooth, weather-driven temporal dynamics in line with typical diurnal and seasonal patterns. No additional preprocessing such as detrending, normalization, or outlier filtering was applied, in order to preserve the simplicity and robustness of the framework and ensure its direct applicability to operational data. The only transformation performed was the construction of sliding windows to capture temporal dependencies, where each observation consists of a sequence of past values $X_i = [x_i, x_{i+1}, \dots, x_{i+W-1}]$. This windowing enhances the ability of the model to learn patterns over time without requiring complex preprocessing. Formally, each input sequence aggregates the last 48 hourly samples with a stride of one hour. This two-day window provides sufficient temporal context while preserving real-time feasibility. The window size was empirically selected within $[24, 96]$h as the optimal compromise between responsiveness and redundancy. Shorter windows underfit diurnal persistence, whereas longer ones introduce redundant correlation.

4.2. Feature engineering

Sine and cosine coding [37] were applied to capture cyclical temporal features using two harmonics defined by: $\sin_T(t) = \sin\left(2\pi\frac{t}{T}\right)$, $\cos_T(t) = \cos\left(2\pi\frac{t}{T}\right)$; where t represents the hour of the day (0 to 23) and T is 24 for the daily cycle. This preserves the cyclical nature for the model. In this study, these cyclic encodings are used solely as additional inputs to the ELM model. Their continuous trigonometric form allows the network to capture daily and more generally seasonal periodicity without any prior detrending, normalization, or categorical expansion, thus avoiding heavy preprocessing. Concerning missing data and the cleaning process, often due to monitoring or IT issues, a linear interpolation was operated filling the gaps with the average of neighboring values. The longest data gap lasted only a few hours; therefore, the gap-filling process does not significantly alter the dataset. With these preprocessing steps, including sliding windows, temporal encoding, and robust missing data handling, the dataset is ready for simulations and to propose robust conclusions. The first five years of data constitute the in-sample training period, while the sixth year is reserved as a strictly out-of-sample test set. This chronological split guarantees that the testing phase occurs entirely after training, preventing any temporal leakage.

To quantify the link between energy variables, the Mutual Information is suggested (MI; [38]), which measures the dependency between two variables, capturing both linear and non-linear relationships, unlike the Pearson correlation, which only detects linear associations. It is defined as:

$$\mathrm{MI}(X;Y) = \int\int p(x,y)\log\frac{p(x,y)}{p(x)p(y)}dxdy, \tag{1}$$

where $p(x,y)$ is the joint probability density, and $p(x), p(y)$ are the marginal distributions. MI can also be interpreted as the Kullback-Leibler divergence between the joint distribution and the product of the marginals, that is, a measure of how far the variables are from being statistically independent. MI is always non-negative, bounded by $0 \le \mathrm{MI}(X;Y) \le \min(\mathrm{H}(X), \mathrm{H}(Y))$, where $\mathrm{H}(X)$ and $\mathrm{H}(Y)$ are the entropies of X and Y. To ensure comparability, MI is normalized:

$$\mathrm{MI}_{\mathrm{norm}}(X;Y) = \frac{\mathrm{MI}(X;Y)}{\sqrt{\mathrm{H}(X)\mathrm{H}(Y)}}, \quad 0 \le \mathrm{MI}_{\mathrm{norm}}(X;Y) \le 1. \tag{2}$$

The analysis highlights strong dependencies: thermal and bioenergy (0.535), hydropower and thermal (0.373), and solar PV with bioenergy (0.468). Total production is mainly influenced by thermal (0.256), hydro (0.129), imports (0.231), and solar PV (0.124). These relationships hint at a structured interaction between energy sources, reinforcing the relevance of a Multi-Input Multi-Output (MIMO) approach for improved forecasting. Typically, a normalized Mutual Information ($\mathrm{MI}_{\mathrm{norm}}$) above 0.2-0.3 indicates a notable relationship, while values exceeding 0.5 denote strong dependence.

4.3. ELM in prediction context

The Extreme Learning Machine (ELM) is a type of feedforward neural network, that is simple and very effective. Unlike traditional networks that loop and keep sending information from one side to the other through layers of complexity (backpropagation), ELM stands out for its simplicity. It is a direct feedback neural network, with fast learning: a single step, no unnecessary movements, no loss of precision. While a conventional MLP (multilayer perceptron) network is theoretically less efficient than an ELM, a deep learning network (such as LSTM; see Appendix B) will be more optimal than an ELM, but will also be much harder to train. For the ELM, a few initializations and linear regressions suffice, making it eligible for "continuous" training over a sliding window. w is chosen using a brute-force method that tests all possible configurations ($w = 24 : 96$ in 24 h steps), keeping the number of hidden nodes constant (2^{12}). In this context, $w = 48$ remains the best option. This choice is consistent with the need to include at least one full diurnal cycle when modelling non-deseasonalized energy time series, and the addition of simple cyclic encodings (sine–cosine representations of hour-of-day and day-of-year) further stabilizes the forecasts. ELM is composed of three parts: an input layer, a hidden layer and an output layer. The network only needs I input features to become operational. The hidden layer contains H neurons, their number being selected according to the task to be accomplished; it is advisable to propose enough to accomplish the work, to match the weight of the task. From there, O output neurons await the result, ready to compute predictions, which are robust and effective. Below are the main parameters of the ELM forecasting tool using the Mean Squared Error (MSE) minimized across all output variables without additional weighting:

- **Input Layer**: The input layer receives the feature vector (dim $= d$) $\mathbf{x} = [x_1, x_2, \dots, x_d]$;
- **Hidden Layer**: The hidden layer applies a linear combination of inputs followed by a ReLU activation function $h_j(\mathbf{x}) = \max(0, \mathbf{w}_j^T \mathbf{x} + b_j)$, where $\mathbf{w}_j$ is the weight vector and b_j the bias for the jth hidden neuron. The hidden layer output is $\mathcal{H} = [h_1(\mathbf{x}), \dots, h_H(\mathbf{x})]^T$ (H is the number of hidden nodes);
- **Output Layer**: The final prediction is a linear combination of hidden layer outputs $\mathbf{y} = \mathcal{H}^T \mathcal{W}$, where $\mathcal{W}$ is the matrix of output weights;
- **Training Process**: ELM training consists of two phases: random weight initialization between the input and hidden layers, and output weight calculation $\mathcal{W} = \mathcal{H}^+ \mathbf{Y}$, where $\mathbf{Y}$ is the target output and $\mathcal{H}^+$ is the Moore-Penrose pseudo-inverse of $\mathcal{H}$;
- **Multiple Initializations**: To ensure robustness, multiple random weight initializations are performed (50 in the case of this study). The best configuration, determined by evaluating a performance metric such as RMSE ($= \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$) computed in-sample ($N$ the number of samples in the evaluation set), is selected ($\hat{}$ for prediction).

In Appendix B, the schema of the multivariate ELM used in the simulations is presented, with $7 \times 48 + 2$ inputs and 7 outputs. The inputs correspond to all electrical variables (7), spanning 2 days and two time indices $\sin_T(t)$ and $\cos_T(t)$, while the outputs represent each of the 7 electrical variables at the forecast horizon. For validation, a persistence model, a simple and reliable reference, is used. It is the measure against which the predictions of the ELM model are compared. In the absence of complex patterns, it intervenes, assuming that the future closely resembles the last known moment. Known by some as naive forecasting, it considers each subsequent step as an echo of the previous one.

Mathematically, the forecast, $\hat{y}_{t+h}$, for a future point $t + h$, is defined by the relation $\hat{y}_{t+h} = y_t$, where y_t marks the last value recorded at time t, and h is the deviation projected into the future. Its simple logic proves particularly effective when past numbers have an influence on what will follow. In the multi-output case (MIMO − ELM), the target matrix $\mathbf{Y}$ has one column per energy source. The closed-form solution $\mathcal{W} = \mathcal{H}^{\dagger}\mathbf{Y}$ then simultaneously yields the output weights for all variables, defining a unified regression framework. This formulation makes explicit that the MIMO extension is not a duplication of single-output models, but an integrated approach ensuring consistency across outputs. For forecasting the energy mix, purely data-driven approaches often remain the only practical option, since physics-based or knowledge-driven models are not yet available at this scale, due to unstable installed capacity, unplanned outages, maintenance operations, and strong variability across seasons.

4.4. Evaluation metrics

To evaluate the performance of the ELM forecasting model, key error metrics are used: RMSE, MAE, and R^2, along with their normalized versions. These metrics provide a comprehensive assessment of accuracy and facilitate comparisons with benchmark models. RMSE measures prediction errors by emphasizing larger deviations, making it sensitive to outliers. MAE, defined as $\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$, captures the average absolute error and treats all deviations equally, offering robustness against extreme values. R^2 quantifies the variance and ranges from 0 (no explained variance) to 1 (perfect fit). To ensure comparability across datasets, normalized error metrics are used ($\bar{}$ for mean):

$$\mathrm{nRMSE} = \frac{\mathrm{RMSE}}{\bar{y}}, \quad \mathrm{nMAE} = \frac{\mathrm{MAE}}{\bar{y}}. \tag{3}$$

Performance is benchmarked against a persistence model using the gain metric (or skill score):

$$\mathrm{G} = \frac{\mathrm{RMSE}_{\mathrm{persistence}} - \mathrm{RMSE}_{\mathrm{ELM}}}{\mathrm{RMSE}_{\mathrm{persistence}}}, \tag{4}$$

where a positive G indicates superior ELM performance. For instance, if the persistence model yields RMSE = 0.50 and ELM achieves 0.45, then G = 0.1, reflecting a 10% error reduction.

5. Results and discussion

All codes and datasets used in this study are publicly available at the following repository: https://github.com/cyrilvoyant/Data_Viz_2024. For the simulations, 80% of the data was used for in-sample training, while the remaining 20% was allocated for out-of-sample testing (error metrics were calculated exclusively on the data not involved in the learning phase). The separation between sample sets is chronological: the first values are used for training, and the subsequent ones for testing (hindcast process).

In the following, "electricity consumption" refers to the total amount produced by all energy sources, disregarding losses (even though in some cases these may reach a few percent). To ensure a fair comparison across models, identical datasets, forecasting horizons, input variables, sliding-window procedures, and chronological train/test splits (80/20) were used for MIMO-ELM, SISO-ELM, and LSTM. All models were evaluated using the same performance metrics, including absolute error measures (RMSE, MAE and their normalized counterparts), the coefficient of determination (R^2), and the gain G, interpreted as a skill score relative to the persistence benchmark. This combination provides a balanced view of accuracy, explanatory power, and relative improvement.

In addition, results are reported for TimeGPT, a large pre-trained transformer-based model for time-series forecasting, used here as a modern reference baseline. In contrast to the other approaches, TimeGPT is

evaluated in a zero-shot configuration, without local training or fine-tuning, reflecting a realistic deployment scenario where predictions are obtained through an external inference service. Although its learning paradigm differs from that of the locally trained models, the same input data, forecasting horizons, and evaluation metrics are used to ensure a consistent and transparent comparison.

To complement error-based metrics, a Wilcoxon signed-rank test is applied to paired hourly errors on the test set. Additional metrics are intentionally avoided to preserve clarity and reproducibility, as the selected indicators are those most commonly adopted in energy and solar forecasting studies.

5.1. Feasibility

In Fig. 1, the prediction results for the six energy sources and the total electrical consumption of Corsica are presented. Error metrics are computed over 1 year of hourly predictions for a forecasting horizon ranging from 1 to 10 hours. These curves confirm that the forecasting method is valid, both visually and in terms of the error metrics for each plot, demonstrating its efficiency. For instance, on the first day (between 8:00 and 22:00), the two peaks in Total production ≈ Total consumption (morning and evening) are attributed to solar and thermal energy, respectively. The cross-pattern of the method enhances its responsiveness. The contribution of bio-energy to the energy mix is minimal, with low variability, suggesting that a simple persistence model is more efficient due to the negative gain value. This behaviour reflects the fact that, despite its low variance, the bioenergy series presents almost no exploitable temporal structure and behaves as a noise-dominated, near piecewise-constant process. In such cases, persistence already lies close to the optimal prediction bound, and regression-based models cannot extract additional information, which explains the negative gain observed.

Wind energy, on the other hand, remains challenging to predict, with an nRMSE equal to 0.4216. Keeping in mind that the wind turbine installed power is only 18 MW spread over three wind farms relatively close geographically. This behaviour is consistent with the statistical properties of the six-year wind record, which exhibits large relative fluctuations around its mean level and marked hour-to-hour intermittency. In addition, the limited installed capacity (18 MW) concentrated across only three geographically close wind farms produces a weak aggregated signal with a low signal-to-noise ratio, making short-term fluctuations proportionally dominant and mechanically limiting the achievable accuracy of deterministic forecasts.

For solar energy, the error metrics are particularly favorable; however, it is important to note that the spatial distribution of solar PV stations reduces resource variability, making the prediction easier. Across all horizons, the conclusions remain consistent, indicating the proposed methodology is efficient. Some examples of predictions using the ELM methodology are available in Appendix C concerning horizons $h+5$ and $h+6$.

The distribution of energy sources at $11am$ and $6pm$ on the third day (Fig. 1) is depicted in Fig. 2. In the evening (when the solar resource is tending to disappear), the distribution is predominantly composed of thermal energy, though this pattern is not consistent across different hours (as observed, for instance, during the first peak in demand where solar PV is more important; This highlights the complexity in forecasting the energy mix and underscores the intricate relationship between time and the various factors captured by the ELM model, which are neither straightforward nor understandable by the human mind.

5.2. Forecasting performance

In the following sections, Tables 1–5 present the metrics obtained, and Fig. 3 plots them for easier visualization. The results presented in Table 1 clearly illustrate the fact that R^2 values obtained for various energy sources are correlated with the different forecasting horizons (h). This observation suggests that the ELM model not only effectively captures the inherent patterns in energy production data but also consistently does so over time. Indeed, with R^2 values close to 1, it becomes evident that the ELM model accounts for a substantial part of the observed variance of the dataset, surpassing traditional persistence methods. In light of these results, it appears that the ELM model is a particularly robust approach for energy output forecasting, offering a nuanced

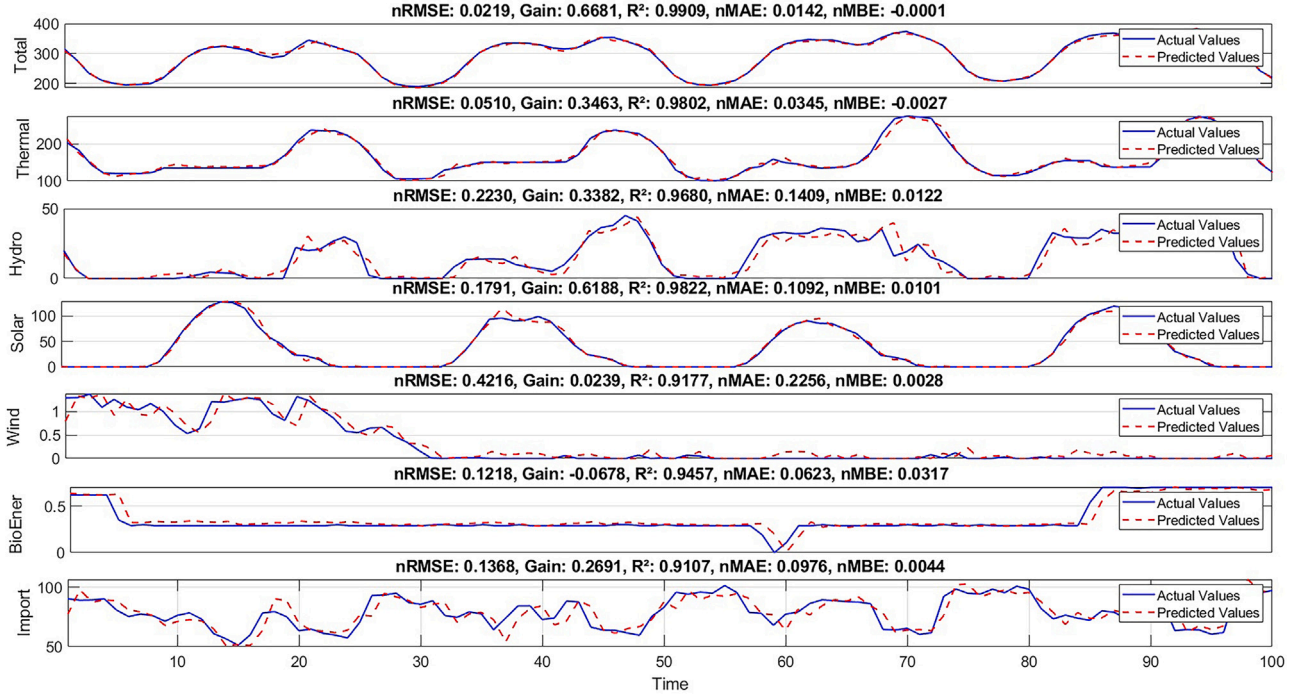
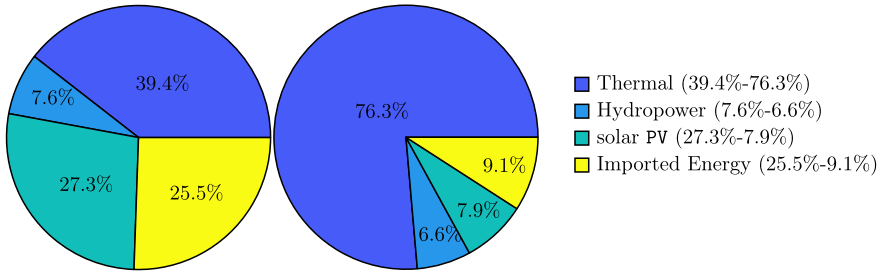


Fig. 1. Profile of the 7 predictions (1 h horizon) for ~ 4 consecutive days (100 h) in Winter. Electrical variables in MW and time in hour.



(a) 11*am* (Total: 335 MW) (b) 6*pm* (Total: 383 MW)

Fig. 2. Breakdown of energy consumption in percentages. Wind and bioenergy not represented ($\leq 0.1\%$).

Table 1
R^2 for energy sources (unitless) computed for 1 year of predictions (out-sample).

Horizon (h)	Total	Thermal	Hydropower	Solar PV	Wind	Bioenergy	Imported
1	0.9909	0.9802	0.9680	0.9822	0.9177	0.9457	0.9107
2	0.9800	0.9528	0.9303	0.9493	0.8155	0.8562	0.8021
3	0.9710	0.9309	0.9073	0.9227	0.7285	0.7635	0.7298
4	0.9646	0.9139	0.8933	0.9033	0.6548	0.6715	0.6822
5	0.9602	0.8999	0.8847	0.8910	0.5915	0.5822	0.6539
6	0.9568	0.8887	0.8786	0.8833	0.5388	0.4971	0.6365
7	0.9544	0.8800	0.8741	0.8794	0.4884	0.4165	0.6256
8	0.9527	0.8732	0.8704	0.8776	0.4452	0.3402	0.6186
9	0.9512	0.8677	0.8676	0.8767	0.4066	0.2687	0.6138
10	0.9502	0.8626	0.8651	0.8764	0.3746	0.2019	0.6096

Table 2
nMAE for energy sources (unitless) computed for 1 year of predictions (out-sample).

Horizon (h)	Total	Thermal	Hydropower	Solar PV	Wind	Bioenergy	Imported
1	0.0142	0.0345	0.1409	0.1092	0.2256	0.0623	0.0976
2	0.0221	0.0570	0.2124	0.1838	0.3562	0.1184	0.1494
3	0.0271	0.0708	0.2490	0.2229	0.4514	0.1644	0.1764
4	0.0303	0.0799	0.2688	0.2461	0.5269	0.2028	0.1918
5	0.0323	0.0867	0.2803	0.2570	0.5877	0.2356	0.2003
6	0.0337	0.0917	0.2882	0.2626	0.6373	0.2639	0.2050
7	0.0347	0.0954	0.2932	0.2636	0.6827	0.2892	0.2076
8	0.0355	0.0980	0.2975	0.2616	0.7215	0.3118	0.2092
9	0.0360	0.1003	0.3008	0.2610	0.7548	0.3319	0.2102
10	0.0365	0.1024	0.3032	0.2610	0.7829	0.3500	0.2113

Table 3
nMBE for energy sources (unitless) computed for 1 year of predictions (out-sample).

Horizon (h)	Total	Thermal	Hydropower	Solar PV	Wind	Bioenergy	Imported
1	−0.0001	−0.0027	0.0122	0.0101	0.0028	0.0317	0.0044
2	−0.0002	−0.0062	0.0206	0.0121	0.0026	0.0668	0.0096
3	−0.0003	−0.0092	0.0258	0.0138	0.0024	0.0972	0.0140
4	−0.0004	−0.0119	0.0290	0.0147	0.0019	0.1243	0.0177
5	−0.0004	−0.0144	0.0315	0.0133	0.0011	0.1483	0.0210
6	−0.0005	−0.0166	0.0334	0.0113	0.0000	0.1695	0.0237
7	−0.0006	−0.0186	0.0344	0.0090	−0.0010	0.1884	0.0260
8	−0.0006	−0.0203	0.0348	0.0071	−0.0017	0.2057	0.0280
9	−0.0007	−0.0217	0.0351	0.0050	−0.0029	0.2213	0.0297
10	−0.0008	−0.0230	0.0356	0.0038	−0.0039	0.2354	0.0311

Table 4
nRMSE for energy sources (unitless) computed for 1 year of predictions (out-sample).

Horizon (h)	Total	Thermal	Hydropower	Solar PV	Wind	Bioenergy	Imported
1	0.0219	0.0510	0.2230	0.1791	0.4216	0.1218	0.1368
2	0.0324	0.0787	0.3292	0.3026	0.6313	0.1981	0.2037
3	0.0389	0.0952	0.3797	0.3734	0.7658	0.2541	0.2380
4	0.0430	0.1062	0.4072	0.4178	0.8635	0.2994	0.2581
5	0.0456	0.1146	0.4233	0.4436	0.9393	0.3377	0.2693
6	0.0476	0.1208	0.4344	0.4589	0.9981	0.3705	0.2760
7	0.0488	0.1254	0.4425	0.4666	1.0513	0.3991	0.2801
8	0.0498	0.1289	0.4488	0.4700	1.0947	0.4244	0.2828
9	0.0505	0.1317	0.4536	0.4716	1.1322	0.4468	0.2845
10	0.0511	0.1342	0.4578	0.4722	1.1623	0.4667	0.2861

understanding of temporal dependencies while significantly reducing the risk of overfitting, as it is often the case with more complex models.

Table 2 presents the nMAE values calculated for different energy sources, thereby providing insights into the prediction accuracy achieved by the ELM model. These low nMAE values, notably for thermal and hydraulic energy, imply that the ELM model predictions are quite close to observed values, underscoring its strong predictive performance for these conventional sources. However, in contrast, higher nMAE values for solar and wind energy are observed. This indicates significant challenges due to the intrinsic variability of these RE sources. Furthermore, bioenergy and imported energy sources exhibit significant variability, underlining the need for improved forecasting models. In summary, although conventional energy sources benefit from relatively reliable forecasts, predictions for RE sources must be refined to better support grid integration and enhance overall stability.

The results in Table 3 show nMBE values close to zero for all energy sources (except for large horizons concerning Bio-energy), signifying that the ELM model provides unbiased forecasts. This is particularly important in energy forecasting, where systematic overestimation or underestimation can lead to significant operational and financial consequences. As a remark, thermal production and consumption are consistently underestimated, while Hydropower, solar PV, and imports are always overestimated.

In Table 4, the nRMSE values, demonstrate the strong performance of the ELM model across different energy sources. The consistently low nRMSE values underscore the model precision in forecasting. Notably, the error in total consumption "Total" column) shows a very low nRMSE (< 0.0511 for $h = 10$), indicating the high potential of ELM for efficient electrical demand management.

5.3. Model comparison with persistence

Table 5 summarizes the G metrics computed over 1 year of predictions using out-of-sample data (distinct from the training set), highlighting the significant performance improvements achieved by ELM over traditional persistence methods. These consistently positive Gain ($G > 0$) for each energy source indicate that ELM significantly improves forecast accuracy. This improvement is particularly valuable because it supports more informed decision-making, improves operational efficiency, and enhances the reliability of the energy supply.

Table 5
G for energy sources (unitless).

Horizon (h)	Total	Thermal	Hydraulic	Solar PV	Wind	Bioenergy	Imported
1	0.6681	0.3463	0.3382	0.6188	0.0239	−0.0678	0.2691
2	0.7306	0.4154	0.4174	0.6530	0.0320	−0.1360	0.3462
3	0.7619	0.4677	0.4738	0.6920	0.0441	−0.1733	0.4042
4	0.7831	0.5056	0.5094	0.7216	0.0559	−0.1985	0.4407
5	0.7980	0.5294	0.5329	0.7450	0.0657	−0.2106	0.4641
6	0.8065	0.5424	0.5459	0.7621	0.0754	−0.2154	0.4790
7	0.8099	0.5454	0.5435	0.7750	0.0834	−0.2178	0.4892
8	0.8086	0.5404	0.5269	0.7841	0.0907	−0.2196	0.4957
9	0.8038	0.5308	0.4979	0.7903	0.0973	−0.2154	0.4975
10	0.7975	0.5197	0.4626	0.7945	0.1033	−0.2089	0.4948

As such, ELM seems to represent a significant advance in energy forecasting, presenting a distinct advantage over conventional approaches. In Fig. 3, the predicted results for each of the six energy sources are presented, as well as the total electricity consumption of Corsica, over a forecast horizon ranging from 1 to 10 hours. G provides valuable information on the performance of ELM over different energy sources and forecast intervals. From this analysis, several key observations are drawn: The gain in total energy forecasting increases from 0.6681 at a 1-hour horizon, peaking at 0.8099 at 7 hours before slightly declining, suggesting that 7 hours is the optimal forecasting horizon. A similar trend is observed for thermal energy, where the gain rises from 0.3463 to 0.5454 at 7 hours, and for hydraulic energy, which improves from 0.3382 to 0.5459 at 6 hours before a minor decrease. For solar energy, the gain consistently increases from 0.6188 to 0.7945 over the entire 10-hour horizon, highlighting strong predictive accuracy or alternatively, emphasizing the poor performance of a persistence-based approach. Wind energy, though lower in gains, shows steady improvement from 0.0239 at 1 hour to 0.1033 at 10 hours. In contrast, bioenergy presents negative gains, starting at −0.0678 and decreasing further to −0.2196 by 8 hours, indicating persistent forecasting challenges. Finally, imported energy sees a steady gain increase from 0.2691 at 1 hour to 0.4975 at 9 hours, where it stabilizes, suggesting sustained improvement in forecast precision. In summary, the ELM provides significant improvements with respect to persistence for most energy sources, especially for total consumption, solar PV, thermal and hydro. However, bioenergy forecasting remains challenging and requires further refinement, even with low installed capacity close to 20 MW for.

5.4. Comparison between ELM and LSTM

One may question whether the Extreme Learning Machine (ELM) is better suited for energy demand and supply forecasting than usual deep learning methods, particularly in the MIMO context. According to the literature, these approaches often outperform simpler models. However, this trend is not always sustainable, as increasing model complexity can reduce interpretability without necessarily improving performance meaningfully. In this section, the simple linear-like ELM model and the deep Long Short-Term Memory (LSTM) approach are compared. In Appendix B, the theoretical complexity differences between these methods, but it is also crucial to evaluate them in a real setting. For fairness, both MIMO − ELM and LSTM were trained and tested under identical experimental conditions: the same dataset, forecasting horizons, input structure, and sliding-window protocol. LSTM hyperparameters (hidden units, learning rate, batch size, epochs) were tuned within realistic computational limits to provide a representative benchmark. The intent is to reflect operationally realistic conditions rather than an exhaustive optimization of LSTM. This configuration is intentionally under-parameterised relative to standard deep-learning practice, as the purpose of this baseline is to illustrate the computational gap with ELM models under identical hardware constraints rather than to achieve full convergence or to serve as a state-of-the-art reference.

To this end, experiments were conducted on a standard workstation equipped with an Intel Core i7-1365U (13th generation) 1.80 GHz processor and 16 GB of RAM, using CPU only without GPU acceleration. This demonstrates that the proposed framework can be executed in real time on common hardware, reinforcing its suitability for operational contexts. In line with open science principles, all codes are publicly available on GitHub (https://github.com/cyrilvoyant/Data_Viz_2024), allowing full reproducibility of the experiments and validation on any hardware platform.

For the ELM model with 1413,120 parameters, training is completed in 104.1 s per initialization. In contrast, the LSTM model (with tanh

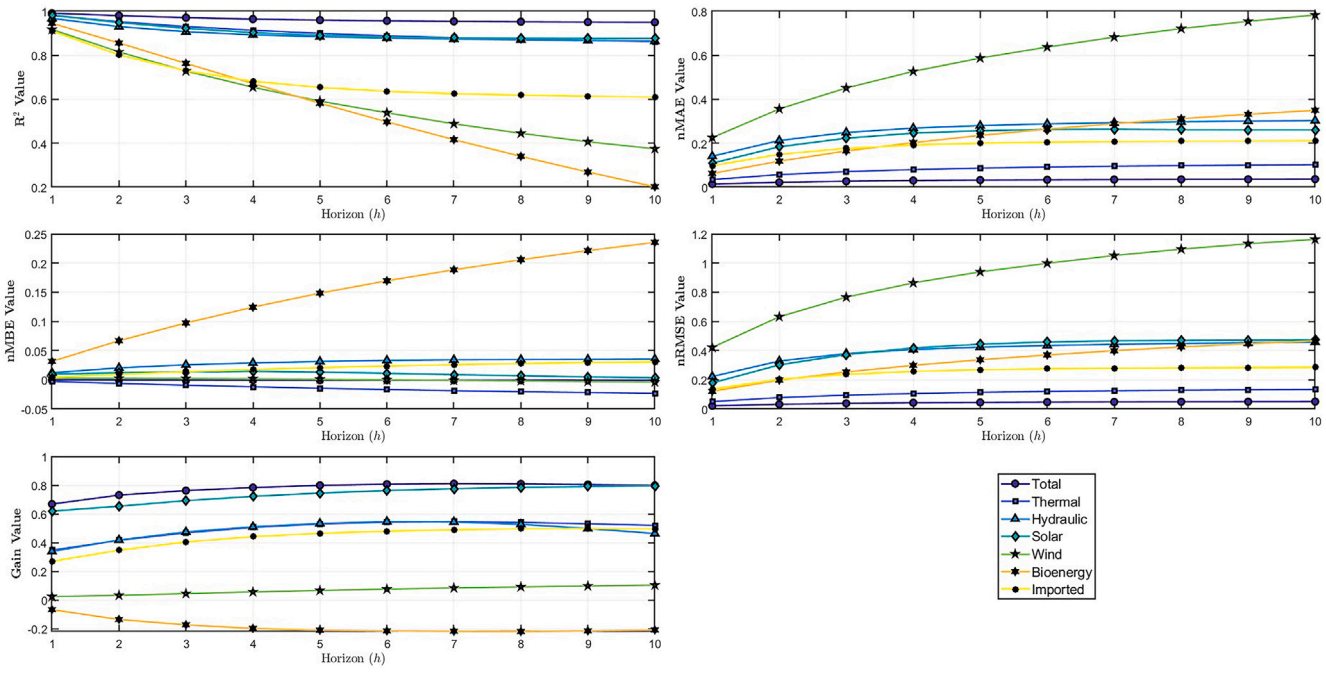


Fig. 3. Error metrics regarding the horizon of prediction.

Table 6
Performance of LSTM for a 5-hour horizon compared to Tables 1–5.

Category	nRMSE	G	nMAE	R^2
Total	0.1722	0.2379	0.1295	0.4336
Thermal	0.3373	−0.3854	0.2593	0.1323
Hydropower	1.1244	−0.2407	0.9957	0.1868
Solar PV	1.3083	0.2478	1.0437	0.0516
Wind	1.5064	−0.4984	1.3293	0.0506
Bioenergy	1.5296	−4.4837	1.4164	0.0571
Imported	0.4664	0.0720	0.3611	0.0879

activation for cell state, sigmoid activation for gates, Adam optimizer, and Mean Squared Error (MSE) loss, trained with MaxEpochs = 5, mini-batch gradient descent, MiniBatchSize = 256, and no explicit early stopping) has 500 hidden neurons and 1023,507 parameters, requiring 2651.6 s for training (approx. 25 times longer). On the same machine, training an LSTM model with 4096 hidden neurons would be infeasible within a reasonable time frame. Although deep learning approaches are often powerful, they are also difficult to interpret (users rarely understand the inner workings of the learning process and mechanisms involved) and demand significant computational resources. ELM, being a linear model, offers a straightforward, closed-form solution (without iterative optimization), while LSTM results may vary depending on the software used (e.g., Python® vs. Matlab® implementations). Moreover, ELM can be deployed in an online learning context, whereas LSTM can be unsuitable for real-time applications. For clarity, results are shown for a representative forecasting horizon (h+5), though similar conclusions hold for other horizons. Table 6 summarizes the performance metrics.

A Wilcoxon signed-rank test applied to paired forecasting errors confirmed that the improvements of ELM over persistence and LSTM are statistically significant for most energy sources (e.g., $p<10^{-6}$ for Total and Solar, $p\approx0.02$ for Wind). These findings underline that ELM provides competitive accuracy with much lower computational requirements than LSTM, reinforcing its operational interest.

5.5. Comparison with a foundational forecasting model (TimeGPT)

To provide a contemporary reference baseline beyond persistence and LSTM, additional results are reported using TimeGPT, a large pre-trained transformer model for time series forecasting. TimeGPT is used in a zero shot setting (no local training), which reflects a realistic deployment scenario where forecasts are generated directly from historical windows via an external inference service. In the present experiments, client side latency remains low (0.21 min for a batched cross validation call), while the underlying model relies on a large server side transformer (on the order of 10^8 parameters). At the aggregate level (mean over all energy sources and horizons in Table 7), TimeGPT achieves nRMSE = 0.409 and nMAE = 0.258, corresponding to a relative gain G ≈ 0.357 compared with the simple persistence baseline (nRMSE = 0.636). These results position TimeGPT as a strong, modern benchmark; however, the proposed MIMO-ELM framework remains preferable in operational settings where model transparency, retraining capability, and multi-variable coherence are required, while maintaining competitive accuracy at negligible computational cost.

Table 7
Aggregate performance (mean over all energy sources and horizons).

Model	$\overline{\text{nRMSE}}$	$\bar{G}$	$\overline{\text{nMAE}}$	$\overline{R^2}$
MIMO-ELM	0.3495	0.4012	0.2285	0.7864
TimeGPT	0.4090	0.3569	0.2580	0.7522

5.6. Comparison between MIMO and SISO

The objective of this section is to quantify the trade-off between computational speed and predictive performance when comparing the MIMO and SISO approaches. The MIMO model leverages cross-variable interactions, potentially improving accuracy, whereas the SISO approach treats each variable independently. If the MIMO model operates in 104.1 s (see Section 5.4), the SISO approach requires a longer execution time (240.7 s for the 7 energy data). This increase is primarily due to the independent training of multiple models for each variable. The results, presented in Table 8, illustrate SISO performance for prediction horizons ranging from 1 hour, passing through 5 hours, up to 10 hours. Both approaches utilize the same ELM architecture and analytical resolution method. However, in the SISO case, a strictly univariate framework is considered (see Appendix B), treating each variable separately without leveraging cross-variable dependencies. This distinction highlights the potential efficiency and accuracy benefits of the MIMO approach. As expected, the prediction error (SISO) increases as the forecast horizon extends. The total energy prediction remains the most stable across all horizons, with only a moderate increase in error. Thermal and hydroelectric energy also show reasonable predictability, although their errors grow significantly after 5 hours. Wind and bioenergy exhibit the highest errors, with, in the first case, an nRMSE of 1.17 at 10 hours, highlighting the challenge of forecasting these highly variable energy sources. Overall, the trend suggests that while the model maintains acceptable accuracy up to 5 hours, performance deteriorates beyond this point, particularly for RE sources. The performance of the SISO and MIMO models was evaluated based on the normalized root-mean-square error (nRMSE) across different forecasting horizons. The results for the MIMO approach are presented in Table 4. Overall, the MIMO approach provides a slight improvement over SISO, particularly for hydropower and solar energy, with error reductions of up to 3.9% and 2.7% at 5 h and 10 h, respectively. The total energy prediction remains stable across both models, while wind energy remains the most challenging variable to forecast, with minimal improvement. For bioenergy, MIMO underperforms, with an increase in nRMSE of up to 4.9% at 5 h, suggesting weaker cross-variable dependency for this source. A Wilcoxon signed-rank test confirmed that these differences between MIMO and SISO are not statistically significant, which is consistent with the modest numerical gaps observed. This outcome is expected for variables such as bioenergy, whose near piecewise-constant behaviour contains neither exploitable

Table 8
Comparison of forecasting errors (nRMSE and nMAE) for SISO approach at different horizons (1 h, 5 h, and 10 h) with relative difference to MIMO (in Parentheses). A positive value indicates higher error in SISO compared to MIMO, meaning MIMO is more accurate.

Variable	nRMSE			nMAE		
	1 h	5 h	10 h	1 h	5 h	10 h
Total	0.0219 (0.0%)	0.0467 (2.4%)	0.0520 (1.8%)	0.0143 (0.7%)	0.0320 (−0.9%)	0.0363 (−0.5%)
Thermal	0.0517 (1.4%)	0.1163 (1.5%)	0.1351 (0.7%)	0.0330 (−4.3%)	0.0859 (−0.9%)	0.1003 (−2.1%)
Hydropower	0.2307 (3.5%)	0.4403 (4.0%)	0.4732 (3.4%)	0.1470 (4.3%)	0.2930 (4.5%)	0.3118 (2.8%)
Solar PV	0.1836 (2.5%)	0.4555 (2.7%)	0.4851 (2.7%)	0.1095 (0.3%)	0.2435 (−5.3%)	0.2463 (−5.6%)
Wind	0.4221 (0.1%)	0.9514 (1.3%)	1.1786 (1.4%)	0.2289 (1.5%)	0.6344 (7.9%)	0.8744 (11.7%)
Bioenergy	0.1172 (−3.8%)	0.3219 (−4.7%)	0.4476 (−4.1%)	0.0543 (−12.8%)	0.2199 (−6.7%)	0.3322 (−5.1%)
Imported	0.1384 (1.2%)	0.2770 (2.9%)	0.2925 (2.2%)	0.0977 (0.1%)	0.2007 (0.2%)	0.2098 (−0.7%)

temporal structure nor meaningful cross-dependencies. In this case, MIMO cannot provide an advantage over SISO, since the shared representation has no informative variability to capture, whereas the formulation still ensures consistency across the other energy components. In conclusion, while MIMO marginally enhances accuracy for most variables, its gains remain moderate. Beyond these numerical results, MIMO forecasting presents specific challenges compared to SISO. It requires handling higher dimensionality and capturing statistical dependencies between energy sources, while ensuring reconciliation between aggregated and disaggregated forecasts. In this study, the difference between the direct prediction of the total and the sum of the predictions of each component remained below 0.2 percentage points. Although the impact on nRMSE was not statistically significant for the present dataset, this internal coherence is guaranteed and represents a key operational advantage. The approach therefore moves from isolated series prediction to a macro-level modeling framework, where both accuracy and consistency must be preserved simultaneously. This highlights the novelty of the present work, since previous studies applied ELM mainly in single-output settings, whereas here it is extended to a multi-output framework ensuring coherent forecasts across multiple energy sources and their aggregate.

6. Conclusion

This study introduces a unified MIMO-ELM framework that successfully addresses the forecasting needs of complex multi-source energy systems. By simultaneously predicting seven grid variables while preserving internal coherence, the approach delivers accurate forecasts with computational efficiency unmatched by deep learning alternatives. Applied to the Corsica insular grid, the model achieves nRMSE below 5% for total consumption up to 10 hours ahead, demonstrating robust performance across diverse energy sources without requiring extensive preprocessing or stationarity assumptions. This study intentionally relies exclusively on endogenous historical power data in order to isolate the structural effects of MIMO versus SISO formulations; incorporating exogenous predictors such as weather or NWP forecasts would shift the analysis toward a different modelling objective and make it difficult to attribute performance differences to the architecture itself.

6.1. Practical implications

The integration of sliding windows and cyclic temporal encodings allows the model to capture seasonal variations and diurnal patterns directly within its structure, eliminating the need for heavy preprocessing. For total energy consumption, the model maintains stable predictions with moderate error growth across horizons. Thermal and hydroelectric generation remain highly predictable, though uncertainty naturally increases beyond five hours. Renewable energy sources, particularly wind (nRMSE = 42% at 1 h), introduce greater volatility due to their inherent variability and relatively limited installed capacity (18 MW across three geographically close wind farms). Solar PV forecasting, while challenging (nRMSE > 40%), benefits from spatial aggregation effects that partially smooth resource variability.

A key methodological contribution lies in the MIMO formulation itself. Unlike traditional SISO approaches that treat each variable independently, MIMO-ELM performs a single closed-form regression yielding simultaneous forecasts for all energy sources and their aggregate. This design ensures internal consistency: the difference between direct total prediction and the sum of individual components remains below 0.2 percentage points throughout testing. While this coherence did not yield statistically significant RMSE improvements over SISO in the present dataset, it represents a critical operational advantage for grid operators who require reconciled forecasts for decision-making. The framework thus transitions from isolated series prediction to macro-level modeling where accuracy and consistency are preserved simultaneously. Beyond predictive accuracy, ELM demonstrates substantial computational advantages over deep learning methods. Given reasonable though non-exhaustive hyperparameter tuning, ELM proved competitive with LSTM while being approximately 25 times faster to train (104 s vs. 2652 seconds on standard hardware). This efficiency stems from its closed-form analytical solution, which avoids iterative optimization entirely. Unlike LSTM, which captures complex temporal dependencies at the cost of high computational burden, limited interpretability, and sensitivity to software implementation, ELM provides transparency and stability suited for real-time applications, including online learning contexts where models must adapt continuously to incoming data. In addition, a comparison with the foundation model TimeGPT was conducted to provide a contemporary reference point beyond classical and deep learning baselines. While TimeGPT benefits from large-scale pretraining and demonstrates strong generalization capabilities, its performance remains comparable to that of the proposed MIMO–ELM framework when evaluated under identical conditions. Importantly, TimeGPT operates as a black-box model and does not support online adaptation or structural coherence across multiple energy variables. As such, although it represents a powerful generic forecasting tool, it does not offer the same level of interpretability, adaptability, or operational control required for realtime multi energy system management.

This work demonstrates that parsimony (selecting the simplest model that performs satisfactorily) delivers not only computational advantages but also operational robustness. The principle holds particular relevance in energy forecasting, where model complexity does not guarantee proportional accuracy gains and may hinder real-time deployment. The MIMO-ELM framework provides grid operators with a practical decision-support tool that balances accuracy, speed, and interpretability. Its closed-form solution, rapid training, and guaranteed internal coherence make it directly applicable to operational contexts without the complexity overhead or opacity of black-box alternatives. Coupled with the integrated multi-output design, this framework delivers forecasts that are coherent, computationally tractable, and immediately deployable; essential qualities as energy systems evolve toward increasingly complex and decentralized architectures.

6.2. Limitations and perspectives

While ELM delivers strong operational performance, it inherits some well known characteristics of single-hidden-layer networks: moderate sensitivity to input scaling, slight variability due to random weight initialization (mitigated here through 50 independent runs), and limited explainability since the model provides an analytical but not fully transparent mapping between inputs and outputs. In addition, the present work does not explicitly address probabilistic uncertainty quantification, sensitivity analyses, or model behavior under extreme or atypical operating conditions (e.g., unusual weather events or demand spikes), which constitute important directions for future research focused on resilience and risk-aware decision-making. Future work could also explore permutation (or SHAP) based sensitivity analyses to enhance interpretability, although their relevance remains limited for highly auto-correlated time-series inputs. Furthermore, the current formulation does not explicitly account for operational constraints such as turbine maintenance schedules, hydropower reservoir levels, or cost-based dispatch priorities. These factors, while outside the scope of purely data-driven forecasting, influence actual grid operations and should be integrated into future decision-support systems to enhance practical applicability.

Three strategic research directions merit investigation. First, incorporating operational constraints and physical limitations would bridge the gap between statistical forecasts and actionable dispatch decisions. Second, extending the MIMO framework to predict multiple horizons simultaneously (rather than training separate models for each horizon) could better capture temporal dependencies between successive time steps, improving overall consistency albeit at higher computational cost. Third, hybrid architectures that combine ELM computational efficiency with deep learning representational power represent a promising avenue

for handling highly volatile renewable sources while maintaining real-time feasibility.

As power systems transition toward decentralized, renewable dominated configurations, the need for transparent and efficient forecasting intensifies. Accurate and rapid predictions enable better energy dispatch optimization, supporting grid operators in balancing economic, technical, and environmental objectives. Beyond traditional grid management, such tools play a vital role in distributed resource coordination, including microgrids where forecasting power limitations and electricity prices over several hours allows optimized energy trading, storage management, and load scheduling.

CRediT authorship contribution statement

Cyril Voyant: Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Milan Despotovic:** Writing – review & editing, Methodology, Data curation. **Luis Garcia-Gutierrez:** Writing – review & editing, Visualization, Methodology. **Mohammed Asloune:** Writing – review & editing, Resources, Data curation. **Yves-Marie Saint-Drenan:** Writing – review & editing, Validation, Formal analysis, Data curation. **Jean-Laurent Duchaud:** Writing – review & editing, Validation, Software, Resources. **Ghjuvan Antone Faggianelli:** Writing – review & editing, Software, Formal analysis, Data curation. **Elena Magliaro:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors wish to express their gratitude to EDF, the French grid manager, for providing the electrical data for Corsica, which was essential for conducting this study. This research was partially funded by the ANR under the grant Fine4CAST project (ANR reference: 22 – PETA – 0008), whose support is gratefully acknowledged.

Appendix A. Workflow and method

The details of the predictive method are given in Fig. A.4. The input data comprises total energy consumption (Tot), as well as subcategories including thermal production (T), hydropower (H), solar PV (S), wind (W), biomass (B), and import (I), alongside transformed temporal indices in sine/cosine form to capture cyclical variations (hours). The preprocessing step divides the time series into overlapping chunks using a sliding window of 48 steps, where each window advances by a single time step, thus generating a multivariate data point at each shift. The model applies a ReLU (rectified linear unit) activation, defined by $\mathcal{H} = \max(0, \mathrm{X}\,\mathcal{W} + b)$, where X is the segmented input, $\mathcal{W}$ is the weight matrix linking input to the hidden layer, and b is the bias term for each neuron. The hidden layer (composed of 4096 neurons) and the output are related by linear regression on the activated hidden state ($\mathcal{H}$), described by $\hat{\mathrm{Y}} = \mathcal{H}\,\mathcal{W}_{out}$, where W_{out} is the optimized output weight matrix derived by least squares ($\mathcal{W}_{out} = (H^{\top}H + \lambda I)^{-1}H^{\top}Y$ with λ the ridge regularization factor). The output layer predicts future values of energy consumption and production $\hat{\mathrm{Tot}}, \hat{\mathrm{T}}, \hat{\mathrm{H}}, \hat{\mathrm{S}}, \hat{\mathrm{W}}, \hat{\mathrm{B}}, \hat{\mathrm{I}}$ at a forecast horizon of h. This architecture optimizes the model by minimizing the root-mean-square error (RMSE) between predictions and observed values. The windowed data approach allows for robust temporal modeling while maintaining granularity over different types of energy production. The model is tested with 50 different initializations to optimize the quality of the forecasts thanks to the regularization of the weights. The number of hidden neurons was selected through systematic exploration using powers of two from 2^7 to 2^{13}; a size of 4096 neurons provided the most stable performance across variables. The regularization parameter was set to $\lambda = 10^{-4}$. Several activation functions (sigmoid, tanh, ReLU) were evaluated and produced comparable forecast accuracy; ReLU was retained due to its simplicity and computational efficiency.

Appendix B. Extreme learning machine (ELM) architecture in MIMO case

The architecture of the ELM model, illustrated in Fig. B.5, follows a straightforward design. The input layer consists of seven variables, each carrying 48 past values, plus two time-based features, totaling 338 inputs. These are passed to a hidden layer with 2^{12} neurons (4096), which extracts meaningful patterns, before being mapped to seven predicted values at the output layer. The number of parameters is structured as follows: the input to hidden layer weights total $338 \times 4096 = 1,384,448$, while the hidden to output layer weights add $4096 \times 7 = 28,672$, leading to a total of 1413,120 parameters. Despite this high parameter count, the ELM model remains computationally efficient. Unlike traditional

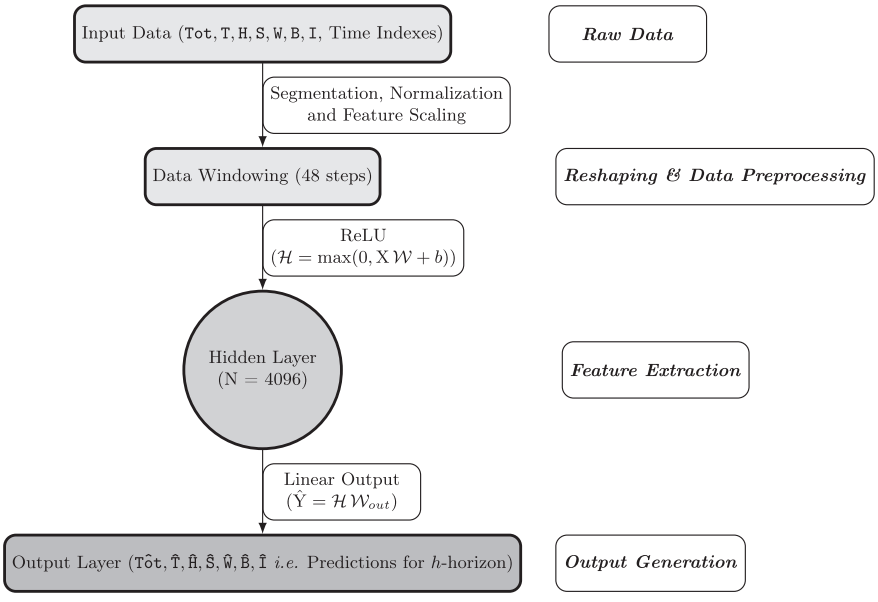


Fig. A.4. Architecture of an extreme learning machine (ELM) used for time series forecasting in the context of multi-source energy regression tasks.

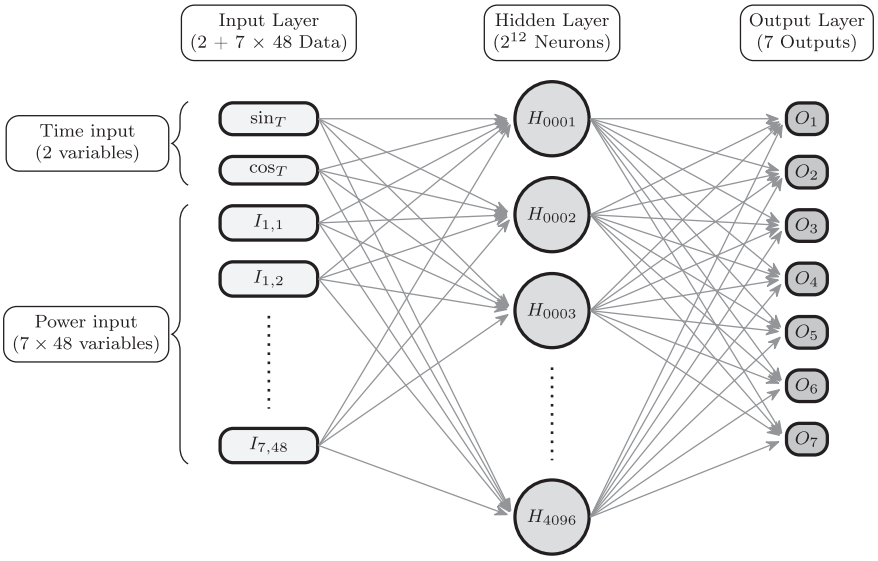


Fig. B.5. Architecture of an extreme learning machine (ELM) for time series forecasting.

deep learning models, it randomly initializes input weights and does not update them, significantly reducing training complexity. The only computational burden lies in computing the pseudo-inverse of the hidden layer output, leading to a training complexity of $\mathcal{O}(H \times n)$, where $H = 4096$ is the number of hidden neurons and n is the number of training samples. The prediction complexity is given by $\mathcal{O}(H \times (I + O))$, where $I = 338$ inputs and $O = 7$ outputs, yielding $\mathcal{O}(1,412,800)$. In contrast, Long Short-Term Memory (LSTM) networks, widely used for time-series forecasting [39], require iterative weight updates, significantly increasing computational cost. Their training complexity is $\mathcal{O}(n \times H^2)$ due to recurrent loops and gating mechanisms, leading to longer training times. Prediction complexity follows $\mathcal{O}(H^2)$, meaning it also scales quadratically with the number of hidden units. To quantify the efficiency difference, the ratio of training complexities is given by $\mathcal{O}_{\text{ELM, train}}/\mathcal{O}_{\text{LSTM, train}} = 1/H$, showing that ELM requires orders of magnitude fewer computations than LSTM ($\approx 10^{-4}$ in this study). Similarly, the prediction complexity ratio is $\mathcal{O}_{\text{ELM, predict}}/\mathcal{O}_{\text{LSTM, predict}} = (I + O)/H$, indicating that ELM is faster at prediction than LSTM, although the advantage is less pronounced compared to training ($\approx 10^{-1}$ in this case). While LSTM models are capable of capturing complex temporal dependencies, their computational cost remains significant due to their recurrent structure and gating mechanisms [40]. In contrast, ELM offers a fast, scalable alternative with minimal training effort and efficient predictions, making it an excellent choice for real-time applications such as energy forecasting and grid management. In addition to architectural considerations, part of the performance gain can be attributed to the structured use of temporal information through sliding windows and time-related encodings. Previous studies have shown that incorporating ordinal and cyclical temporal features can improve short-term forecasting accuracy by approximately 1% in terms of normalized error metrics, particularly in low-variability and operational forecasting contexts [37,41]. In the present framework, this effect is implicitly captured through the sliding-window formulation and the joint learning of multiple correlated energy signals. While modest in magnitude, such improvements are consistent across horizons and contribute meaningfully to the overall robustness and stability of the predictions, especially when aggregated across multiple energy sources.

Appendix C. Some prediction examples

This section presents results for the 5 h and 10 h horizons (Figs. C.6 and C.7). In both cases, there's an issue around the 65-70th hour. When thermal output is underestimated, hydro is overestimated, yet total consumption remains accurate. Forecasting across multiple sources proves challenging; the 5–10 h horizon might reveal the limits of this method and, more broadly, the limits of using time-series methods for such forecasts. From these figures, there is an impression that some anti-correlations are present between errors from thermal and hydro, for example, around the 70th hour. To verify this assumption, a scatter plot of the errors for each part of the energy mix is presented in Fig. C.8. As visible, there is, in fact, no clear feature, making the validation of statistical dependencies between errors challenging.

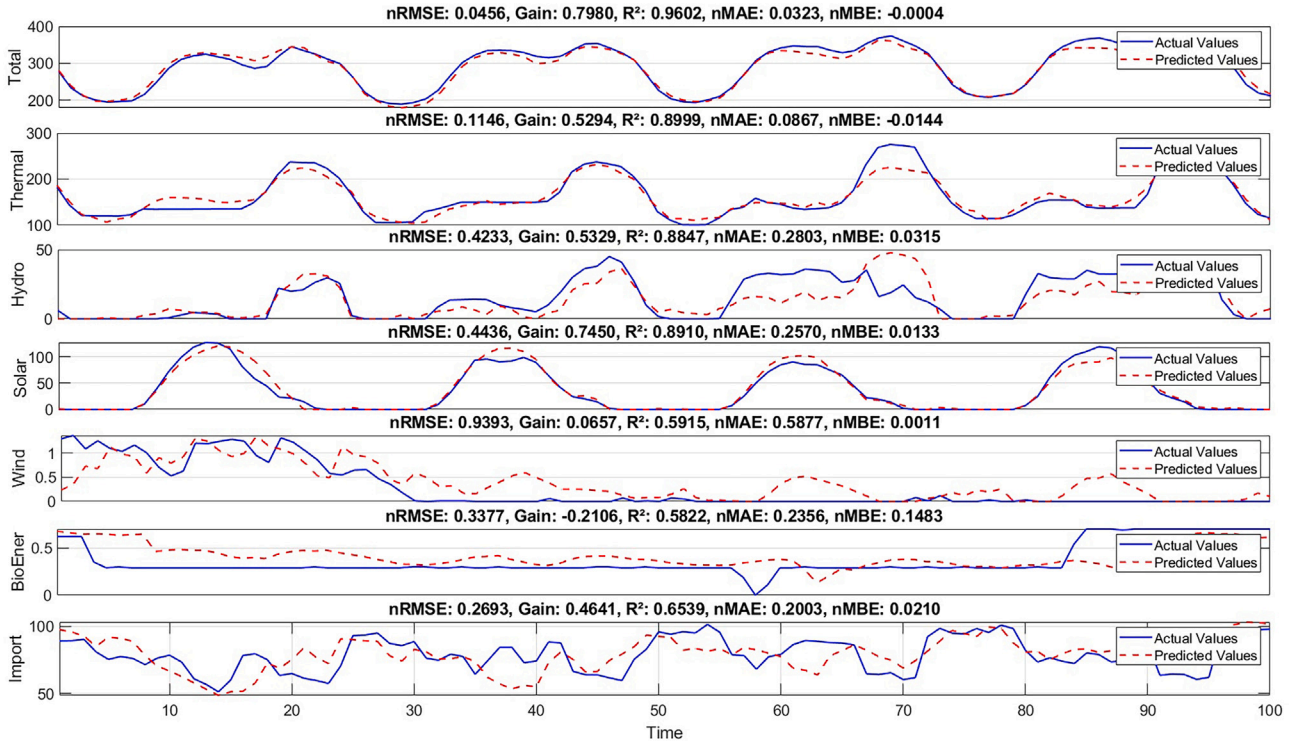


Fig. C.6. Profile of the 7 predictions (5-hour horizon) for ~ 4 consecutive days (100 h) in Winter. Electrical variables in MW and time in hour.

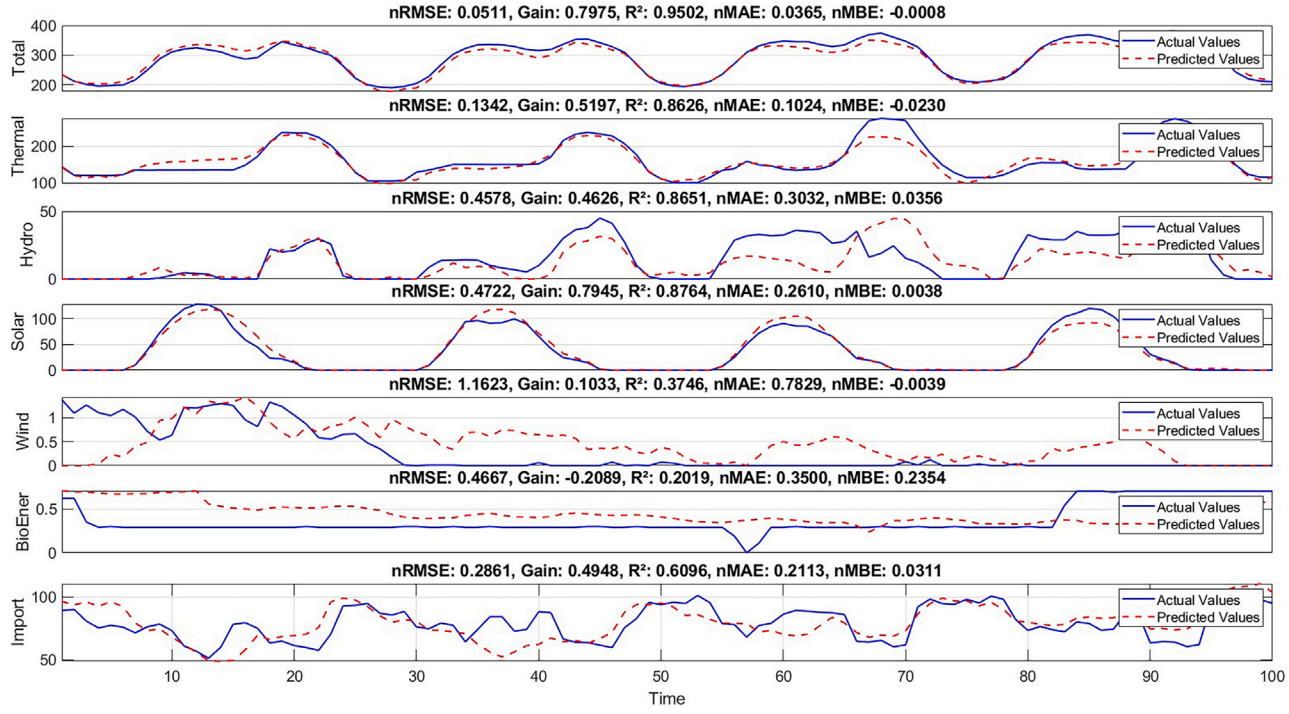


Fig. C.7. Profile of the 7 predictions (10-hour horizon) for ~ 4 consecutive days (100 h) in Winter. Electrical variables in MW and time in hour.

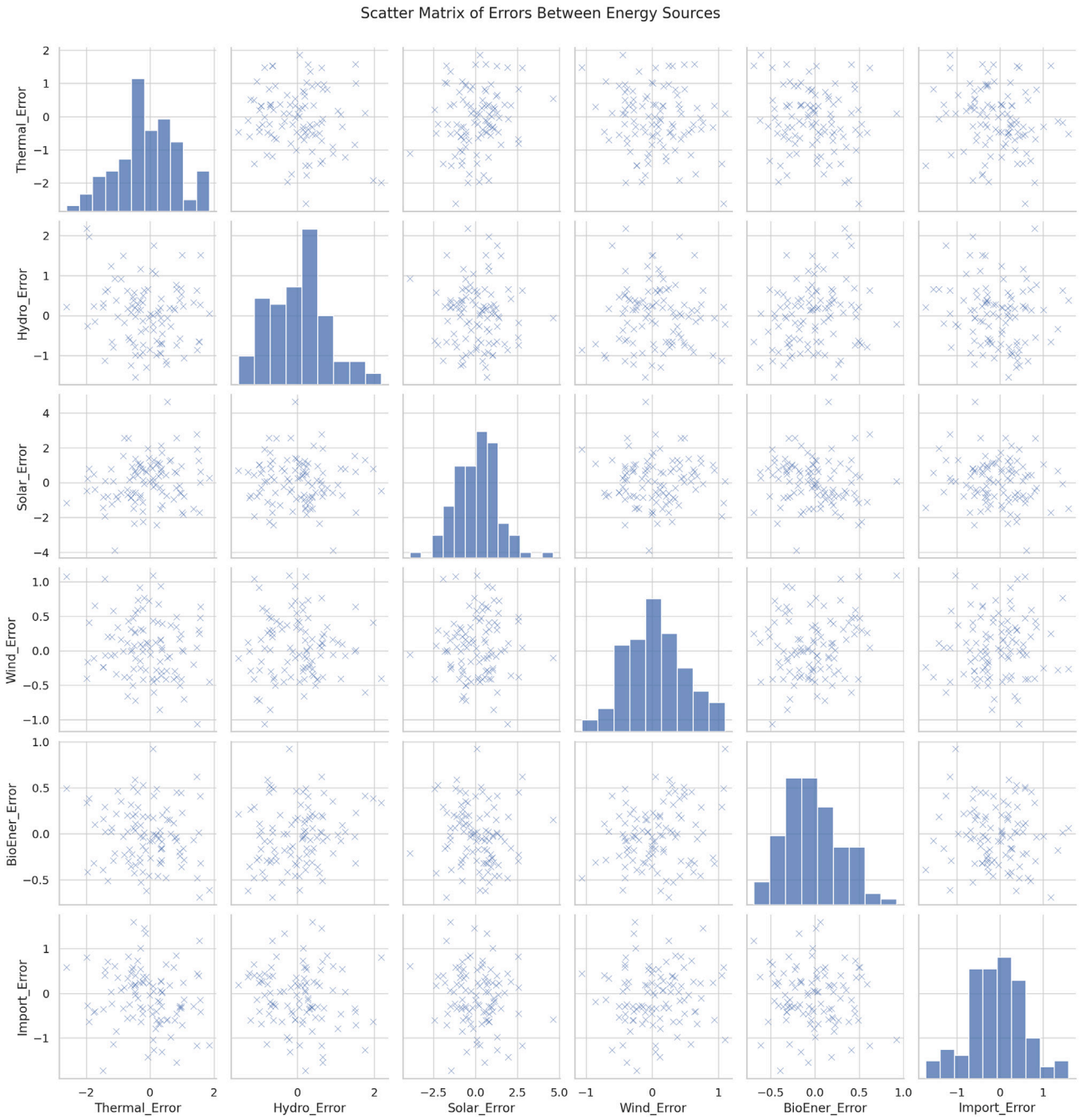


Fig. C.8. Correlation Analysis of forecast errors across energy production sources over the 4 days presented in Figs. C.6 and C.7, with results and trends generalizable to the entire dataset.

Data availability

Data will be made available on request.

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